

# Nuclear Physics Angular Momentum Coupling Guide

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This document provides the theoretical background and practical usage guide for the `angular_momentum_coupling.py` utility. This tool calculates allowed final nuclear states based on quantum mechanical selection rules.

## 1. Theoretical Basis

The code implements the **Channel Spin Coupling Scheme**, which is a standard method for determining angular momentum conservation in nuclear reactions.

### Conservation Laws

In any nuclear reaction  $Target + Particle \rightarrow Final\_State$ , the following quantities are conserved:

#### 1. Total Angular Momentum ( $\vec{J}$ ):

$$\vec{J}_{final} = \vec{J}_{target} + \vec{s}_{particle} + \vec{\ell}$$

#### 2. Parity ( $\pi$ ):

$$\pi_{final} = \pi_{target} \times \pi_{particle} \times (-1)^\ell$$

Where:

- $\vec{J}_{target}$ : Spin of the target nucleus.
- $\vec{s}_{particle}$ : Intrinsic spin of the interacting particle (projectile or transferred cluster).
- $\vec{\ell}$ : Relative orbital angular momentum between the target and the particle.

### Coupling Scheme Used in Code

The code performs the vector addition in two steps (Channel Spin representation):

#### 1. Calculate Channel Spin ( $\vec{S}$ ): Couples the target spin and particle spin.

$$\vec{S} = \vec{J}_{target} + \vec{s}_{particle}$$

Possible values:  $|J_{target} - s_{particle}| \leq S \leq J_{target} + s_{particle}$

2. **Calculate Final Spin** ( $\vec{J}_{final}$ ): Couples the channel spin with the orbital angular momentum.

$$\vec{J}_{final} = \vec{S} + \vec{\ell}$$

Possible values:  $|S - \ell| \leq J_{final} \leq S + \ell$

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## 2. Application to Different Reaction Types

To use the code correctly, you must identify the correct "Particle" input based on the reaction mechanism.

### A. Resonant Capture Reactions

Examples:  $(p, \gamma)$ ,  $(n, \gamma)$

In these reactions, a projectile fuses with the target to form a compound nucleus (resonance).

- **Particle Input:** 1/2+

### B. Single-Nucleon Transfer Reactions

Examples:  $(d, p)$ ,  $(p, d)$ ,  $(^3\text{He}, d)$ ,  $(d, ^3\text{He})$ ,  $(n, d)$ ,  $(d, n)$ ,  $(d, t)$ ,  $(t, d)$ ,  $(\alpha, t)$ ,  $(t, \alpha)$ ,  $(^3\text{He}, \alpha)$ ,  $(\alpha, ^3\text{He})$

In these reactions, a single nucleon is transferred between the projectile and the target. You need to input the properties of the **transferred nucleon**.

- **Particle Input:** 1/2+

**Note on j-transfer:** In transfer physics, we often discuss the transferred total angular momentum  $j$  (where  $\vec{j} = \vec{\ell} + \vec{s}$ ). The code lists results by Channel Spin  $S$ . The set of allowed final  $J^\pi$  values is identical regardless of the coupling order.

### C. Two-Nucleon Transfer Reactions

In these reactions, a pair of nucleons is transferred. The total spin of this pair depends on the coupling of the two nucleons.

#### 1. Identical Nucleon Transfer (2n or 2p)

Reactions:  $(p, t)$ ,  $(t, p)$  (2n transfer);  $(^3\text{He}, n)$  (2p transfer)

For the transfer of two identical nucleons ( $2n$  or  $2p$ ) in the same shell model orbit, the **Pauli Exclusion Principle** dictates that their total wavefunction must be antisymmetric.

- If they are in a relative  $s$ -state (spatially symmetric,  $L = 0$ ), their **spin wavefunction must be antisymmetric**.
- An antisymmetric spin state for two fermions corresponds to a **Singlet State** ( $S = 0$ ).
- **NDS Policy:** For  $(p, t)$ ,  $(t, p)$ , and  $({}^3\text{He}, n)$  reactions, it is standard to assume the transferred pair is in a relative  $s$  state ( $S = 0$ ).
- **Particle Input:**  $0^+$

## 2. Neutron-Proton Transfer ( $1n1p$ )

*Reactions:*  $(\alpha, d)$ ,  $(d, \alpha)$ ,  $({}^3\text{He}, p)$ ,  $(p, {}^3\text{He})$

For the transfer of a neutron-proton pair, the selection rules depend on the projectile/ejectile properties.

- $(\alpha, d)$  **and**  $(d, \alpha)$ :
  - Both  $d$  and  $\alpha$  have  $T = 0$ . Thus, only  $T = 0$  transfer is allowed.
  - Antisymmetry requires  $S + T$  to be odd (for relative  $L = 0$ ). Since  $T = 0$ , the transfer is restricted to  $S = 1$  (spin triplet).
  - **Particle Input:**  $1^+$
- $({}^3\text{He}, p)$  **and**  $(p, {}^3\text{He})$ :
  - Both particles have  $S = 1/2, T = 1/2$ .
  - Allowed transfers are  $(S = 0, T = 1)$  and  $(S = 1, T = 0)$ .
  - **Particle Input:**
    - For  $S = 0$  component:  $0^+$
    - For  $S = 1$  component:  $1^+$
  - *Note: Both components can contribute.*

## D. Cluster Transfer Reactions

*Examples:*  $({}^6\text{Li}, d)$ ,  $({}^7\text{Li}, t)$

An alpha particle is transferred.

- **Particle Input:**  $0+$

## E. Alpha Inelastic Scattering

Examples:  $(\alpha, \alpha')$

- $(\alpha, \alpha')$ : The alpha particle has spin 0. Since the projectile spin cannot flip, the angular momentum transfer is purely orbital ( $\vec{\ell}$ ). This selectively excites **Natural Parity** states ( $\pi = (-1)^\ell$ ).
- **Particle Input:**  $0+$

## 3. Interpreting the Output

When you run the code:

```
python .github/scripts/angular_momentum_coupling.py 3/2- 1/2+
```

You receive output like:

Target:  $J=3/2 \pi=-$  | Particle:  $s=1/2 \pi=+$

Channel Spins S: 1, 2: from  $|3/2 - 1/2|$  to  $3/2 + 1/2$

```
-----
L   Wave  Parity   s      Final Jπ
-----
0   s      -       1      1-
0   s      -       2      2-
-----
1   p      +       1      0+, 1+, 2+
1   p      +       2      1+, 2+, 3+
-----
```

- **L (Wave):** The orbital angular momentum ( $\ell$ ) between target and particle, ranging from 0 to 4 in the output (s, p, d, f, g waves).
- **Parity:** The parity of the final state, determined by  $\pi_{final} = \pi_{target} \times \pi_{particle} \times (-1)^\ell$ .

- **s (Channel Spin):** The intermediate coupling  $S = \vec{J}_{target} + \vec{s}_{particle}$  before adding orbital angular momentum.
  - For  $3/2^-$  target and  $1/2^+$  particle:  $S$  ranges from  $|3/2 - 1/2| = 1$  to  $3/2 + 1/2 = 2$ .
- **Final  $J^\pi$ :** All allowed total angular momentum and parity combinations for each  $(L, S)$  pair, calculated from  $|S - \ell| \leq J_{final} \leq S + \ell$ .

Example Analysis:  $^{59}\text{Cu}(p, \gamma)^{60}\text{Zn}$

- **Target:**  $^{59}\text{Cu}$  ( $3/2^-$ )
- **Particle:** Proton ( $1/2^+$ )
- **Channel Spins:**  $S = 1, 2$  (from  $|3/2 - 1/2|$  to  $3/2 + 1/2$ )
- **Results:**
  - **s-wave capture** ( $L = 0$ ): Parity =  $(-) \times (+) \times (-1)^0 = -$ 
    - $S = 1: J^\pi = 1^-$
    - $S = 2: J^\pi = 2^-$
  - **p-wave capture** ( $L = 1$ ): Parity =  $(-) \times (+) \times (-1)^1 = +$ 
    - $S = 1: J^\pi = 0^+, 1^+, 2^+$  (from  $|1 - 1|$  to  $1 + 1$ )
    - $S = 2: J^\pi = 1^+, 2^+, 3^+$  (from  $|2 - 1|$  to  $2 + 1$ )
  - **d-wave capture** ( $L = 2$ ): Parity =  $-$ 
    - $S = 1: J^\pi = 1^-, 2^-, 3^-$
    - $S = 2: J^\pi = 0^-, 1^-, 2^-, 3^-, 4^-$

## 4. Isospin Coupling

For transfer reactions, isospin ( $T$ ) provides additional selection rules.

- **Single Nucleon Transfer:** Transferring one nucleon ( $n$  or  $p$ ) changes isospin by  $T_{transfer} = 1/2$ .
- **Identical Nucleon Pair (2n/2p):** In a relative  $L = 0$  state, the Pauli Principle requires  $S = 0$  (spin singlet) and  $T = 1$  (isospin triplet). Thus  $T_{transfer} = 1$ .
- **Neutron-Proton Pair (np):**
  - Deuteron-like ( $S = 1, T = 0$ ):  $T_{transfer} = 0$
  - Singlet ( $S = 0, T = 1$ ):  $T_{transfer} = 1$

**Selection Rule:** The final isospin is determined by vector addition:

$$\vec{T}_f = \vec{T}_{target} + \vec{T}_{transfer}$$

Note: This tool calculates angular momentum ( $J^\pi$ ) only. Isospin selection rules must be applied separately.

## 5. Summary of Transferred Particle Properties

| Reaction Mechanism   | Common Examples                                                                    | Physical Process                           | $S_{particle}$ | $T_{particle}$                |
|----------------------|------------------------------------------------------------------------------------|--------------------------------------------|----------------|-------------------------------|
| Radiative Capture    | $(p, \gamma), (n, \gamma)$                                                         | Projectile capture                         | 1/2            | 1/2                           |
|                      | $(\alpha, \gamma)$                                                                 | Projectile capture                         | 0              | 0                             |
| Inelastic Scattering | $(\alpha, \alpha')$                                                                | Excitation (Natural Parity)                | 0              | 0                             |
| One Neutron Transfer | $(d, p), (p, d), (t, d), (d, t), (\alpha, {}^3\text{He}), ({}^3\text{He}, \alpha)$ | Transfer of $n$                            | 1/2            | 1/2                           |
| One Proton Transfer  | $(d, n), (n, d), ({}^3\text{He}, d), (d, {}^3\text{He}), (\alpha, t), (t, \alpha)$ | Transfer of $p$                            | 1/2            | 1/2                           |
| Two-Nucleon Transfer | $(p, t), (t, p), ({}^3\text{He}, n)$                                               | Transfer of $2n$ or $2p$ ( $L_{rel} = 0$ ) | 0              | 1                             |
|                      | $(\alpha, d), (d, \alpha)$                                                         | Transfer of $np$ (Deuteron-like)           | 1              | 0                             |
|                      | $({}^3\text{He}, p), (p, {}^3\text{He})$                                           | Transfer of $np$ (Mixed)                   | 0 and 1        | 1 ( $S = 0$ ) / 0 ( $S = 1$ ) |
| Cluster Transfer     | $({}^6\text{Li}, d)$                                                               | Transfer of $\alpha$ -cluster              | 0              | 0                             |